

AFARA, Algeria

WMO: 606770

Lat: 25.333N

Lon: 7.800E

Elev: 4068

StdP: 12.66

Time Zone: 1.00 (E01)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 30.8	(c) 34.0	(d) -7.3	(e) 4.3	(f) 54.3	(g) -3.1	(h) 5.4	(i) 57.0	(j) 18.6	(k) 68.4	(l) 13.4	(m) 63.9	(n) 0.1	(o) N/A	(p) 0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	101.4	59.7	100.1	59.5	98.8	59.4	64.7	83.2	63.4	85.7	62.3	87.5	1.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	89.3	71.1	56.0	77.8	73.4	52.3	67.7	74.7	32.0	81.2	30.9	84.7	30.1	86.8	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.2	(b) 12.7	(c) 9.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 16.2	(b) 12.7	(c) 9.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	72.1	51.2	55.9	64.0	74.4	82.3	87.6	88.3	86.6	83.1	73.8	63.0	54.0
	DBStd	14.15	5.24	7.11	6.88	5.60	4.50	3.15	2.73	2.79	3.60	4.42	4.90	5.14
	HDD50	88	45	21	2	0	0	0	0	0	0	0	0	20
	HDD65	1240	428	268	103	3	0	0	0	0	0	1	94	343
	CDD50	8146	82	184	437	732	1000	1128	1186	1133	992	739	390	143
	CDD65	3825	1	12	73	285	535	678	721	668	542	275	34	1
	CDH74	57800	26	226	1152	3801	7681	11014	11850	10293	7809	3244	604	100
	CDH80	32966	0	51	338	1712	4331	7021	7614	6212	4352	1261	69	5
Wind	WSAvg	1.0	0.6	0.3	0.5	2.0	2.1	1.8	0.0	0.2	0.9	1.5	1.4	1.1
Precipitation	PrecAvg	1.7	0.2	0.1	0.2	0.1	0.1	0.2	0.0	0.2	0.2	0.2	0.1	0.1
	PrecMax	3.3	0.8	0.5	0.8	0.4	0.9	0.9	0.3	0.9	1.0	1.5	1.0	0.6
	PrecMin	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.6	0.2	0.1	0.2	0.1	0.2	0.2	0.1	0.2	0.2	0.3	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.7	85.7	89.6	97.4	100.5	103.5	102.7	101.4	98.4	93.6	84.3	80.6
		MCWB	49.6	52.2	53.9	57.1	59.2	60.1	60.0	60.5	59.7	58.2	54.8	50.7
	2%	DB	73.7	80.4	86.7	93.9	98.7	101.8	101.0	99.0	97.0	91.2	82.1	77.2
		MCWB	47.2	50.4	52.5	56.4	58.9	59.6	59.7	60.6	59.3	57.9	53.7	49.7
	5%	DB	70.2	76.1	84.0	91.2	96.9	100.5	99.8	97.7	95.8	89.1	80.1	73.0
		MCWB	45.8	48.2	51.9	55.3	58.6	59.3	59.5	60.4	59.1	57.3	52.6	47.7
	10%	DB	67.2	72.7	80.8	88.8	95.1	99.0	98.5	96.3	94.3	86.9	77.9	69.3
		MCWB	44.5	47.2	51.2	54.4	57.9	59.1	59.4	60.1	58.9	56.4	51.7	46.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	54.0	57.7	60.6	63.0	63.6	64.3	67.7	66.2	62.4	57.9	52.9
		MCDB	69.8	77.3	74.0	85.2	84.9	91.3	86.2	78.0	75.9	73.1	79.7	72.2
	2%	WB	49.0	52.0	55.8	58.5	61.5	61.8	62.3	65.3	64.5	61.0	55.2	50.3
		MCDB	69.4	73.2	75.1	84.6	87.7	92.1	90.7	83.6	78.4	78.1	78.1	73.8
	5%	WB	46.4	50.3	54.1	57.2	60.4	60.9	61.2	63.7	63.0	59.9	53.7	48.7
		MCDB	68.9	69.6	76.2	86.7	89.4	94.3	93.6	88.0	82.5	79.7	75.9	71.6
	10%	WB	44.8	48.2	52.2	55.8	59.4	60.2	60.3	62.7	61.9	58.7	52.5	46.9
		MCDB	66.9	70.7	78.8	84.9	89.9	94.6	94.5	89.4	84.2	80.7	74.6	68.5
Mean Daily Temperature Range	5% DB	MDBR	31.1	28.6	27.3	25.6	23.5	22.1	20.3	20.4	22.2	24.3	28.3	29.6
		MCDBR	34.6	32.6	31.9	27.6	25.3	24.1	22.8	22.1	23.6	25.8	30.0	34.7
		MCWBR	18.3	15.7	14.2	10.9	9.2	8.1	8.0	7.1	7.7	8.7	12.8	17.6
	5% WB	MCDBR	32.1	27.7	25.7	23.3	22.8	21.6	20.1	20.9	21.3	22.1	26.4	32.6
		MCWBR	17.4	14.0	10.8	8.6	7.7	6.6	6.6	6.0	5.4	6.5	11.2	16.7
Clear-Sky Solar Irradiance	taub	0.293	0.328	0.427	0.506	0.581	0.595	0.476	0.497	0.506	0.453	0.363	0.309	
	taud	2.426	2.310	1.969	1.793	1.658	1.646	1.917	1.884	1.855	1.974	2.209	2.397	
	Ebn at Noon	307	302	275	256	236	231	260	255	249	256	277	296	
	Edh at Noon	32	39	58	70	81	81	62	63	64	54	40	32	
All-Sky Solar Radiation	RadAvg	1497	1739	2087	2240	2286	2368	2505	2290	2059	1776	1524	1380	
	RadStd	66	95	124	107	101	123	74	98	97	110	57	63	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page